MADHU VADLAMUDI

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SUMMARY

- Accomplished **Data Engineer** with 5+ years of experience in designing, building, and maintaining scalable big data solutions across diverse industry domains.
- Expertise in **Hadoop ecosystems** (Cloudera, Hortonworks) and core components including **HDFS, YARN, MapReduce, Hive, Kafka**, and **HBase** for distributed data processing.
- Advanced proficiency in **Apache Spark** and **PySpark** for high-performance, in-memory analytics and large-scale batch processing.
- Strong background in building **ETL pipelines** using **Apache Flume**, **Apache Airflow**, **Microsoft SSIS**, and **Pentaho Data Integration**, supporting data ingestion from diverse sources.
- Skilled in integrating **NoSQL databases** (e.g., HBase, MongoDB) with traditional **RDBMS** to enable real-time analytics and hybrid data architectures.
- Proficient in **HiveQL, Spark SQL, and Python**, enabling the creation of reusable, modular data transformation workflows.
- Experienced in **CI/CD pipeline development**, automating data workflow deployments for improved agility and reliability.
- Hands-on with **Git and GitHub** for version control, code collaboration, and managing data engineering projects in **cloud-native environments**.
- Adept at **cloud infrastructure monitoring** and performance tuning to support scalable and fault-tolerant **data platforms** on AWS and Azure.
- Demonstrated experience in designing and implementing **RESTful APIs** for exposing data services and enabling seamless system integration.
- Knowledgeable in **Integrated Communication Systems and Information Management (ICSIM)** to streamline data flow and enhance decision-making processes across distributed systems.

TECHNICAL SKILLS

- **Big Data & Distributed Systems:** HDFS, Hive, Pig, Sqoop, YARN, Apache Spark (Core, SQL, Streaming), Kafka, PySpark, Spark SQL
- **Cloud Platforms & Services:** AWS (S3, Redshift, EMR, Lambda, EC2, Glue, Athena), Azure (Data Factory, Synapse, Blob Storage, Azure ML Studio), GCP (BigQuery, Dataflow, Cloud Storage, GCP ML APIs)
- **Programming & Scripting Languages:** Python, SQL, Scala, Shell Scripting, PySpark
- **ETL & Data Integration Tools:** IBM InfoSphere DataStage (v8.x/v9.1), SSIS, Talend, Autosys, Apache NiFi
- **Data Warehousing & Databases:** Snowflake, Teradata, Oracle 9i/10g, DB2, SQL Server, MySQL, Google BigQuery
- **Machine Learning & Analytics:** Linear Regression, Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, SVM, K-Means, KNN, PCA, LDA, Naive Bayes, AdaBoost, NLP (NLTK, SpaCy, Gensim), TensorFlow, Keras, AWS ML, Azure ML
- **Business Intelligence & Reporting Tools:** Power BI, Tableau, Excel (Advanced), Google Analytics
- **Version Control & DevOps:** Git, GitHub, Agile Methodologies, CI/CD Pipelines
- **Development Tools & IDEs:** PyCharm, Jupyter Notebook, Spyder, Visual Studio Code
- **Data Governance & Security:** Data Privacy Compliance (HIPAA, GDPR), Role-Based Access Control, Encryption, Data Masking
- **Additional Expertise:** Data Modeling, Data Quality Management, Performance Optimization, Customer-Centric Analytics

EXPERIENCE

Verizon Wireless Systems — **Jan 2023 – Current**

Data Engineer

- Designed and deployed scalable data integration pipelines using Databricks and Azure Data Factory, processing over 5TB of data daily across structured and semi-structured formats.
- Engineered data solutions using Google Cloud Platform (GCP), leveraging BigQuery, Cloud Functions, and Dataflow for efficient and scalable processing.
- Implemented data governance strategies focusing on metadata management, data quality, and observability within big data pipelines and lakehouse architectures on GCP.
- Optimized complex ETL workflows by tuning Spark jobs and Delta Lake tables, reducing data processing time by 40%.
- Partnered with data science teams to operationalize ML models in real-time production pipelines using Python and GCP services.

- Built automated data validation and monitoring frameworks, increasing data accuracy and trustworthiness by 25%.
- Led the transition from on-premise data warehousing to Snowflake, achieving 50% cost savings and significantly faster query performance.
- Designed, tested, and deployed relational database solutions, including optimized stored procedures, views, and triggers using MySQL and SQL Server.
- Developed a RESTful API using Python for real-time revenue tracking and analytics, integrating seamlessly with downstream reporting systems.
- Used MySQL Workbench and Birst BI to tune database performance, reducing query time by 25% and expediting dashboard refresh rates by 20%.
- Applied predictive modeling and statistical techniques to support data lifecycle management across relational and big data environments.
- Executed complex cloud data migration strategies to GCP, and managed high-availability cloud-native data platforms.
- Built interactive dashboards using Looker and Data Studio, enabling real-time insights through automated reporting solutions.
- Performed robust data analysis using Python (NumPy, SciPy, Pandas, Matplotlib, StatsModels) and R, supporting advanced analytics, feature engineering, and exploratory data insights.

Coforge **Feb 2018 – July 2021**
Data Engineer

- Conducted big data requirement analysis to design scalable ETL pipelines and business intelligence solutions, aligning closely with stakeholder needs.
- Built 3NF-compliant data models for Operational Data Stores (ODS) and OLTP systems, and developed dimensional models using Star and Snowflake Schemas for analytical workloads.
- Utilized Snowflake to eliminate data redundancy and integrate real-time streaming data into HDFS from diverse sources using Apache Kafka.
- Designed a centralized enterprise data warehouse model in Snowflake, integrating over 100 datasets to enable unified analytics.
- Managed ingestion and transformation of structured and semi-structured data using Python and AWS S3, and facilitated on-prem to AWS cloud migration of big data workloads.
- Developed and optimized Hive-based ETL processes on Amazon EMR, ensuring data pipelines met complex business logic and SLAs.
- Executed data migration from traditional RDBMS systems to Hadoop using Sqoop, evaluating performance benchmarks and storage efficiency.
- Developed MapReduce programs for data validation and filtration, improving data quality and processing accuracy prior to Hive table ingestion.
- Designed, configured, and maintained a high-throughput Kafka cluster, capable of processing up to 1 million messages per second, using Kafka Producer API 0.6.3.
- Engineered real-time data pipelines using Kafka and Spark Streaming, transforming data into RDDs and DataFrames, and storing outputs in Parquet format in HDFS for efficient querying.

EDUCATION

- **Master of Science: Computer and Information Science**
  Southern Arkansas University - Magnolia, Arkansas AR
- **Bachelor of Technology: Computer Science and Engineering**
  JNTU-KAKINADA - India